Lecture 4
Association Rule Mining

CSC1104: Data Warehousing and Data Mining
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Overview
Association Rules
Generating Association Rules
The Apriori Algorithm
Generating Itemsets
Generating Rules
Measures of Interest: Lift and Leverage

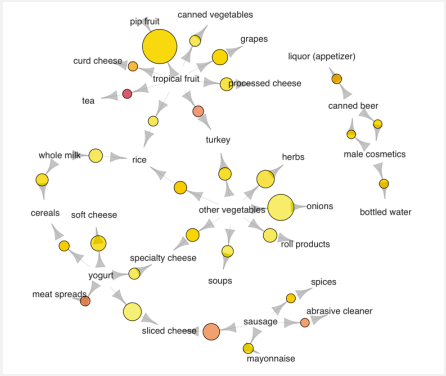
Agenda

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2. Association Rules
   - Generating Association Rules
3. The Apriori Algorithm
4. Generating Itemsets
5. Generating Rules
   - Measures of Interest: Lift and Leverage



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Association Rule Mining



Association Rule Mining is the analysis of data for *patterns* or *co-occurrences* in a dataset.



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Discovering Association Rules

- *Market Basket Analysis* a special form of **Association Rule Mining**.
- The rules generated for Market Basket Analysis are all of a certain restricted kind.
- Aim: identify rules that relate the purchases made by customers in a shop, frequently a large store with many thousands of products, as opposed to those that predict the purchase of one particular item.
- However, these methods are not restricted to the retail industry, but also analysis of items purchased by credit card, patient's medical records, crime data and data from satellites.



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Transactions and Itemsets



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- We will assume that we have a database comprising *n* transactions each of which is *a set of items*.
- Consider each transaction as corresponding to a group of purchases made by a customer, for example *{milk,cheese, bread}* or *{fish, cheese, bread, milk, sugar}*.
- Here *milk, cheese, bread* are **items** and *{milk, cheese, bread}* an **itemset**.
- The aim is finding rules known as **association rules** that apply to purchases made: buying *fish and sugar* is often associated with buying *milk and cheese*.
- However, we want rules that meet certain criteria for being *interesting*.

Search Criteria



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- Including an item in a transaction just means that some quantity of it was bought.
- We *do not record* quantity or the items that a customer did *not* buy.
- We only look for rules that link all the items that were actually bought.

Search Criteria



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- Assume that there are *m* possible items that can be bought with the letter *I* to denote the set of all possible items.
- The value of *m* (all combinations) can be very large.
- It partly depends on whether to consider all meat sold as a single item *meat* or as a separate item for each type of meat *(beef, lamb, chicken etc.)* or as a separate item for each *type and weight combination*.
- Clearly the number of different items that could be considered in a basket analysis is potentially very large.

Ordering Itemsets



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- While order is not necessary, we write a transaction as {cheese, fish, meat}, not {meat, fish, cheese} etc.
- This does no harm, as the meaning is obviously the same, but has the effect of greatly reducing and simplifying the calculations required to discover all the *interesting* rules that can be extracted from the database.

Sample Transactions



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If a database comprises 8 transactions ($n = 8$) and there are 5 different items (unrealistically low), denoted by *a, b, c, d* and *e*, we have $m = 5$ and $I = \{a, b, c, d, e\}$:

Transaction number	Transactions (itemsets)
1	{a, b, c}
2	{a, b, c, d, e}
3	{b}
4	{c, d, e}
5	{c}
6	{b, c, d}
7	{c, d, e}
8	{c, e}

Figure 1: Sample Database of Transactions

All itemsets are subsets of *I*. We do not count the empty set so an itemset can have anything from 1 up to *m* members.

Itemset Count



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- We use the term *support count* of an itemset *S*, or just the **count** of an itemset *S*, to mean the number of transactions in the database matched by *S*.
- We say that an itemset *S* matches a transaction *T* (itself an itemset) if *S* is a subset of *T*, i.e. all items in S are also in T.
- For example, itemset *{bread, milk}* matches the transaction *{cheese, bread, fish, milk, wine}*.
- If an itemset *S = {bread, milk}* has a count of 12, written as:
  `count(S)` = 12 or `count({bread, milk})` = 12, it means that 12 of the transactions in the database contain both the items *bread* and *milk*.
- For now, *n* is not considered.

Itemset Count Examples



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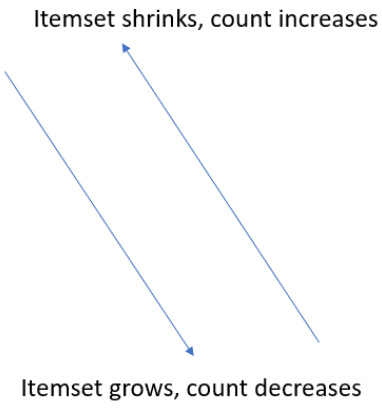
- Count(a) = 2
- Count(a,b) = 2
- Count(a,b,c) = 2
- Count(a,b,c,d) = 1
- Count(a,b,c,d,e) = 1

Itemset Count Examples



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- Count(a) = 2
- Count(a,b) = 2
- Count(a,b,c) = 2
- Count(a,b,c,d) = 1
- Count(a,b,c,d,e) = 1



Support for an Itemset

- **Definition**: The **support** of an itemset S, (`support(S)`), is the *proportion* of itemsets in the database that are matched by S, i.e. the proportion of transactions that contain all the items in S.
- Alternatively: the **frequency** with which the items in S occur together in the database.
- Thus, where n is the number of transactions in the database:

$$support(S) = count(S)/n \quad (1)$$



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Itemset Support Examples

- Support(a) = 2/8 = 0.25
- Support(a,b) = 2/8 = 0.25
- Support(a,b,c) = 2/8 = 0.25
- Support(a,b,c,d) = 1/8 = 0.125
- Support(a,b,c,d,e) = 1/8 = 0.125



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Prediction

- The goal of Association Rule Mining (ARM) is to find rules, known as *association rules*, in the data.
- For example, when items c and d are bought, item e is also purchased.
- Write this as the rule: $cd \rightarrow e$(*meaning* : *cd implies e*).
- Do not interpret this as meaning that buying c and d somehow *causes* e to be bought.
- Think of rules in terms of *predictions*: if we know that c and d were bought we can *predict* that e was also bought.



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Rules and Rulesets

- The rule $cd \rightarrow e$ is typical of many of the rules used in Association Rule Mining in that it is *invariably* incorrect.
- The rule is satisfied for transactions 2, 4 and 7 in Figure 1, but *not* for transaction 6: it is satisfied in 75% of cases.
- For basket analysis, interpreted this as: if bread and milk are bought, then cheese is also bought in 75% of cases.
- Note that the presence of items c, d and e in transactions 2, 4, and 7 can also be used to justify other rules such as
  $c \rightarrow ed$ *and* $e \rightarrow cd$
  which again do not have to be always correct.



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Rules have a Left-hand side and a Right-hand side

- The number of rules generated from a small database is potentially very large, with many having no value.
- A method is required to determine which rules to discard and which to retain.
- If the set of items appearing on the left and right sides of a rule are *L* and *R*, and the rule is written: **L** $\rightarrow$ *R*.
- *L* and *R* must each have *at least* one member and the sets must be disjoint, i.e. no common members.



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ARM Terminology and Basics

- With $L \rightarrow R$, the left and right sides are both sets.
- The **union** of the sets *L* and *R* is the set of items that occur in *either L* or *R*.
- It is written $L \cup R$.
- As *L* and *R* are disjoint and each has at least one member, the number of items in the itemset L $\cup$ R (cardinality), must be at least two.



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Calculating Support for a Ruleset

- For the rule *cd* $\rightarrow$ *e* we have *L* = *{c, d}*, *R* = *{e}* and $L \cup R$ = *{c, d, e}*.
- Count the number of transactions in the database matching the *L* and *R* itemsets.
- Itemset *L* matches four transactions (2,4,6,7) and itemset
  $L \cup R$ matches 3 transactions (2,4,7) so:
  `count(L)` = 4 and `count(L ∪ R)` = 3.
- As there are 8 transactions in the database we can calculate:
  `support(L)` = `count(L)/8` = 4/8 and
  `support(L ∪ R)` = `count(L ∪ R)/8` = 3/8



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Support: Which is higher?

Calculating Support for a Ruleset

- As there are 8 transactions in the database we can calculate:
  `support(L)` = `count(L)/8` = 4/8 and
  `support(L ∪ R)` = `count(L ∪ R)/8` = 3/8



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Interesting Rules

- A large number of rules can be generated from a small database but only use those that satisfy criteria for being *interesting*.
- Never use rules that apply to a *small proportion* of the database: they predict poorly.
- In measuring *interestingness*, the two most common metrics are **support** and **confidence**.



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Support for Rules

- The **support** for a rule $L \rightarrow R$ is the *proportion* of the database to which the rule successfully applies.
- In other words, the proportion of transactions where both items in L and R occur together.
- This value is just the support for itemset $L \cup R$, so we define support for a rule:

$$support(L \rightarrow R) = support(L \cup R) \quad (2)$$



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Predictive Accuracy (Confidence for Itemsets)

- The predictive accuracy of the rule $L \rightarrow R$ is measured by its **confidence**, defined as *the proportion of transactions* for which the rule is satisfied.
- Calculate as:
  the number of transactions matched by the left-hand and right-hand sides combined,
  as a proportion of the number of transactions matched by the left-hand side alone.
- **Confidence** is:

$$count(L \cup R)/count(L) \quad (3)$$



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Confidence for Rules

- Ideally, every transaction matched by L is also matched by $L \cup R$, meaning the value of confidence is 1 and the rule would be called *exact*, i.e. always correct.
- In practice, rules are generally not exact, in which case `count(L ∪ R) < count(L)` and confidence < 1.
- Using the previous slide, we can now define:

$$confidence(L \rightarrow R) = count(L \cup R)/count(L) \quad (4)$$

or exactly the same:

$$confidence(L \rightarrow R) = support(L \cup R)/support(L) \quad (5)$$



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Rule Thresholds

- It is customary to reject any rule for which the support is below a minimum threshold value called **minsup**, typically 0.01 (i.e. 1% or 1 transaction in every 100).
- Also reject any rule with confidence below a minimum threshold value called **minconf**, typically 0.8 (i.e. 80%).
- For the rule $cd \rightarrow e$, the confidence is: `count(c, d, e)/count(c, d)` which is 3/4 = 0.75.
- Using slide 9, can you specify a rule which meets the *minconf* threshold?



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Generating Associating Rules

- The term **supported itemset** is used to denote any itemset for which the value of support $\geq$ **minsup**.
- The terms **frequent itemset** and *large itemset* are often used instead of supported itemset.
- A basic but very inefficient method has two stages.
  1. Generate all supported itemsets $L \cup R$ with cardinality at least two.
  2. For each such itemset generate all the possible rules with at least one item on each side and retain those for which confidence $\geq$ `minconf`.



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Generating Association Rules: issues

- The main problem is with *step 1* is generating all possible itemsets of cardinality two or greater.
- The number of such itemsets depends on the total number of items *m*.
- For a practical application, this can be hundreds or even thousands.



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Get rid of itemsets with size ≤ 1

- The number of possible itemsets $L \cup R$ is the same as the number of possible subsets of *I*, the set of all items, which has cardinality *m*.
- There are 2^m subsets (all combinations of elements).
- Of these, *m* itemsets have a single element and one has no element (the empty set).
- Thus, the number of itemsets $L \cup R$ with cardinality at least 2 is 2^m - m (single element sets) - 1 (empty set).



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Summary

- If m takes the (unrealistically small) value of 20, the number of itemsets $L \cup R$ is 2^{20} - 20 - 1 = 1,048,555.
- If m has a (still small!) value of 100, the number of itemsets $L \cup R$ is 2^{100} - 101, which is approximately 10^{30}.
- Generating all the possible itemsets $L \cup R$ and then checking *every rule* to determine which are supported is unrealistic.
- An optimisation method is required ...



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The Apriori Algorithm





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Theorem: Downward Closure

If an itemset is supported, then all of its (non-empty) subsets are also supported.

Proof

- Removing one or more items from an itemset *cannot* reduce (will often increase) the number of transactions that it matches.
- Thus, the *support* for a **subset of an itemset** must be at least as great as support for the original itemset.
- It follows that any (non-empty) subset of a supported itemset must also be supported.

Note: Revisit slides 11 and 14 for an empirical proof: as dataset decreases, support increases!



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Further Exploitation of this Property

- This result is sometimes called the *downward closure property* of itemsets.
- If we write the set containing all the supported itemsets with cardinality k as L_k, then a second important result follows in **Theorem 2**.
(The use of the letter L stands for *large itemsets*).



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Theorem: Testing for Empty Set

If $L_k = \Theta$ (the empty set) then L_{k+1}, L_{k+2} etc. must also be empty.

Proof

- If supported itemsets of cardinality *k+1* or larger exist, they will have subsets of cardinality *k* and it follows from **Theorem 1** that all of these must be supported.
- However, if $L_k = \Theta$, there are no supported itemsets of cardinality *k* as L_k is empty.
- Hence, there are no supported subsets of cardinality *k+1* or larger and L_{k+1}, L_{k+2} etc. must all be empty.



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Empty Set Optimisations

- Generate the supported itemsets in ascending order of cardinality, those with 1 element, then those with 2 elements, then those with 3 elements etc.
- At each iteration, the set L_k of supported items with cardinality *k* is generated from the previous set L_{k-1}.
- The benefit is that when $L_k = \Theta$ (the empty set), we know that L_{k+1}, L_{k+2} etc. must also be empty.
- Itemsets of cardinality *k+1* or greater are guaranteed *not* to be supported and the algorithm terminates.



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Optimisation Explained

Assume minconf=70%

support (a) = 90%	k=1
support (a, b) = 90%	k=2
support (a, b, c) = 70%	k=3
support (a, b, c, d) = 65%	k=4
support (a, b, c, d, e) = 50%	k=5
support (a, b, c, d, e, f) = 35%	k=6
.....	
.....	
support (a, b, c, d...,x, y, z) = ?	



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Apriori Step 1

- We need a method of going from each set L_{k-1} to the next L_k in turn. There are 2 steps
  - First, we use L_{k-1} to form a candidate set C_k containing itemsets of cardinality *k*.
  - C_k must be constructed in such a way that it is certain to include **all** supported itemsets of cardinality *k*.
  - It may also contain unsupported itemsets: that's ok!



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Apriori Step 2

- Next, generate L_k as a subset of C_k.
- It is possible to discard some members of C_k as possible members of L_k by inspecting the members of L_{k-1}.
- The remainder must be checked against the transactions in the database to establish their support values.
- Only those itemsets with support greater than or equal to `minsup` are copied from C_k into L_k .
- This gives us the **Apriori algorithm** for generating all supported itemsets of cardinality at least 2.



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Apriori Algorithm

```
Create L1 (supported itemsets of cardinality 1)
Set k to 2
while (Lk−1 ≠ Θ)
    Create Ck from Lk−1
    Prune supported itemsets in Ck to create Lk
    k=k+1
```

The set of all supported itemsets is L1 $\cup$ *L2* $\cup \ldots \cup$ *Lk*



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Initial Step

- To begin: construct C_1 (the set of all itemsets comprising just a single item) and make a pass through the database counting the number of transactions that match each of these itemsets.
- Dividing each of these counts by the number of transactions in the database gives the value of **support** for each single-element itemset.
- Discard all those with *support* $<$ *minsup* to give L_1.
- This process can be represented as Figure 2, continuing until L_k is empty.



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Apriori Illustrated

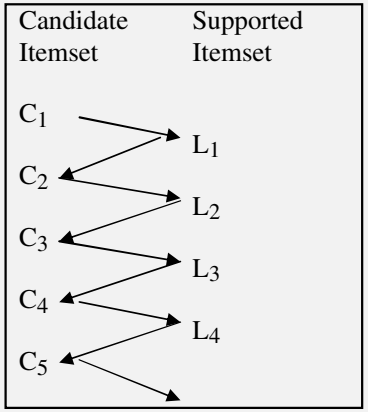


Figure 2: Apriori: C_2 to C_5 generate L_2 to L_5



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Apriori-gen Example (L_4)

- The **Apriori-gen** version takes L_{k-1} and generates C_k without using any of the earlier sets L_{k-1} etc.
- There are two stages, shown on the next slide.
- To illustrate the method, let us assume that L_4 is a list containing 17 itemsets of cardinality 4.

{p, q, r, s}, {r, s, w, x}, {r, v, x, y},
{p, q, r, t}, {r, s, w, z}, {r, v, x, z},
{p, q, r, z}, {r, t, v, x}, {r, v, y, z},
{p, q, s, z}, {r, t, v, z}, {r, x, y, z},
{p, r, s, z}, {r, t, x, z}, {t, v, x, z},
{q, r, s, z}, {v, x, y, z}



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Apriori-Gen Algorithm

Generate C_k from L_{k-1}:

```
Join Step:
In L_{k-1}, compare each member A with every other member B
For all comparisons:
   If A=B for leftmost (k-2) items (all except rightmost)
      put A ∪ B into C_k

Prune Step:
For each c in C_k
   Examine all subsets of c with k-1 elements
   if any subset NOT in L_{k-1}
            delete c from C_k
```

Step 1 (Join) creates the Candidate Set *L* and Step 2 (Prune) creates the Supported Set *C*.



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Join Step(*k=5, k-2=3*)

- From potentially many subsets in C_4, only 6 pairs of elements have the first 3 in common.
- List these together showing the combinations that places their merge into C_5.
- Compare $\{a_1,a_2,a_3,a_4\}$ with $\{b_1,b_2,b_3,b_4\}$: if $(a_1 = b_1)$ AND $(a_2 = b_2)$ AND $(a_3 = b_3)$ then CREATE $\{a_1,a_2,a_3,a_4,b_4\}$

First itemset	Second itemset	Contribution to C_5
$\{p,q,r,s\}$	$\{p,q,r,t\}$	$\{p,q,r,s,t\}$
$\{p,q,r,s\}$	$\{p,q,r,z\}$	$\{p,q,r,s,z\}$
$\{p,q,r,t\}$	$\{p,q,r,z\}$	$\{p,q,r,t,z\}$
$\{r,s,w,x\}$	$\{r,s,w,z\}$	$\{r,s,w,x,z\}$
$\{r,t,v,x\}$	$\{r,t,v,z\}$	$\{r,t,v,x,z\}$
$\{r,v,x,y\}$	$\{r,v,x,z\}$	$\{r,v,x,y,z\}$

Figure 3: Initial Version of candidate set C_5



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Prune Step

- Here, each of the subsets of cardinality four of the itemsets in C_5 are examined in turn, with the following results.

Itemset in C_5	Subsets all in L_4?
$\{p,q,r,s,t\}$	No, e.g. $\{p,q,s,t\}$ is not a member of L_4
$\{p,q,r,s,z\}$	Yes
$\{p,q,r,t,z\}$	No, e.g. $\{p,q,t,z\}$ is not a member of L_4
$\{r,s,w,x,z\}$	No, e.g. $\{r,s,x,z\}$ is not a member of L_4
$\{r,t,v,x,z\}$	Yes
$\{r,v,x,y,z\}$	Yes

Figure 4: Final Version of candidate set C_5



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Final Step

- We can eliminate the first, third and fourth itemsets from C_5, making the final version of candidate set C_5
- {{p, q, r, s, z}, {r, t, v, x, z}, {r, v, x, y, z}}
- The three itemsets in C_5 now need to be checked against the database to establish which are supported.



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Generating *Supported* Itemsets: An Example

- Assume that we have a database with 100 items and a large number of transactions.
- We begin by constructing C_1, the set of itemsets with a single member.
- We make a pass though the database to establish the support count for each of the 100 itemsets in C_1 and from these calculate L_1, the set of supported itemsets that comprise just a single member.
- Assume L_1 has 8 of these members:
  {a}, {b}, {c}, {d}, {e}, {f}, {g}, {h}.
- We cannot generate rules as they have only one element, but we can form *candidate itemsets* of cardinality two.



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Pass 1

- In generating C_2 from L_1, all pairs of (single-item) itemsets in L_1 are considered to match at the join step, since there is nothing to the left of the rightmost element of each one that might fail to match.
- In this case, the candidate generation algorithm gives us as members of C_2 all the itemsets with two members drawn from the eight items a, b, c, . . . h.
- Note that it would be pointless for a candidate itemset of two elements to include any of the other 92 items from the original set of 100, e.g. *{a, z}*, as one of its subsets would be *{z}*, which is not supported.



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Generating C_2

There are 28 possible itemsets of cardinality 2 that can be formed from the items a, b, c, . . . , h:
{a, b}, {a, c}, {a, d}, {a, e}, {a, f}, {a, g}, {a, h},
{b, c}, {b, d}, {b, e}, {b, f}, {b, g}, {b, h},
{c, d}, {c, e}, {c, f}, {c, g}, {c, h},
{d, e}, {d, f}, {d, g}, {d, h},
{e, f}, {e, g}, {e, h},
{f, g}, {f, h},
{g, h}.



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Pass 2

- We now need to make a second pass through the database to find the support counts of each of these itemsets,
- Then, divide each of the counts by the number of transactions in the database and reject any itemsets that have support less than `minsup`.
- Assume in this case that only 6 of the 28 itemsets with two elements turn out to be supported.
- L_2 = {{a, c}, {a, d}, {a, h}, {c, g}, {c, h}, {g, h}}.
- The algorithm for generating C_3 now gives just four members, i.e. {a, c, d}, {a, c, h}, {a, d, h} and {c, g, h}.



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Pass 2 Analysis

- We now check whether each of the candidates meets the condition that all its subsets are supported.
- Itemsets {a, c, d} and {a, d, h} fail this test, because their subsets {c, d} and {d, h} are not members of L_2.
- That leaves just {a, c, h} and {c, g, h} as possible members of L_3.
- A third pass is needed to find the support counts for itemsets {a, c, h} and {c, g, h}.



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Pass 3

- We will assume they both turn out to be supported: L_3 = {{a, c, h}, {c, g, h}}.
- We now need to calculate C_4.
- It has no members, as the two members of L_3 do not have their first two elements in common.
- As C_4 is empty, L_4 must also be empty, which implies that L_5, L_6 etc. must also be empty and the process ends.



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Pass 3 Analysis

- We have found all the itemsets of cardinality at least 2 with just three passes through the database.
- In doing so, we needed to find the support counts for just 100+28+2 = 130 itemsets.
- This is a considerable improvement on checking through the total number of possible itemsets for 100 items, which is approximately 10^{30}.



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Overall Analysis

- The set of all supported itemsets with at least two members is the union of L_2 and L_3:
- {{a, c}, {a, d}, {a, h}, {c, g}, {c, h}, {g, h}, {a, c, h}, {c, g, h}}.
- It has eight itemsets as members.
- Now generate the candidate rules from each of these and determine which of them have a confidence value greater than or equal to `minconf`.



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Analysis: Closure

- An itemset X is **closed** in a data set S, if there exists no proper super-itemset Y such that Y has the *same support count* as X in S.
- Example . . . a=75% a,b=75% : a is *not* closed as (a,b) is a superset of (a)
- An itemset X is a **closed frequent itemset** in set S, if X is both *closed* and *frequent* in S.
- An itemset X is a **maximal frequent itemset** (or max-itemset) in set S, if X is frequent, and there exists no super-itemset Y such that $X \subset Y$ and Y is frequent in S.



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Closure Example

- Let C be the set of *closed* frequent itemsets for a data set S satisfying a minimum support threshold, **minsup**.
- Let M be the set of *maximal* frequent itemsets for S satisfying **minsup**.
- Assume we know the support count of each itemset in C and M.
- C and its count information can be used to derive the entire set of frequent itemsets.
- C contains complete information regarding its corresponding frequent itemsets; while M registers only the support of the maximal itemsets.



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10 itemsets and their supports

Table 1: Begin by sorting on supports

A	100%	A	100%
B	75%	B	75%
C	75%	C	75%
D	50%	AB	75%
AB	75%	AC	75%
AC	75%	D	50%
AD	50%	AD	50%
ABC	50%	ABC	50%
ABD	25%	ABD	25%
BCD	25%	BCD	25%
	Original		Sorted



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Determine Closure



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Table 2: Ask: have you a super itemset elsewhere with $\geq$ support?

closed	A	100%
	B	75%
	C	75%
closed	AB	75%
closed	AC	75%
	D	50%
closed	AD	50%
closed	ABC	50%
closed	ABD	25%
closed	BCD	25%

Determine Maximal Itemsets



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Goal: further reduce itemsets!

Table 3: Supports are not required in search for maximal.

maximal	ABC	Largest must be maximal
maximal	ABD	Largest must be maximal
maximal	BCD	Largest must be maximal
	AB	contained in ABC
	AC	contained in ABC
	AD	contained in ABD
	A	contained in . . .
	D	contained in . . .

Generating Rules

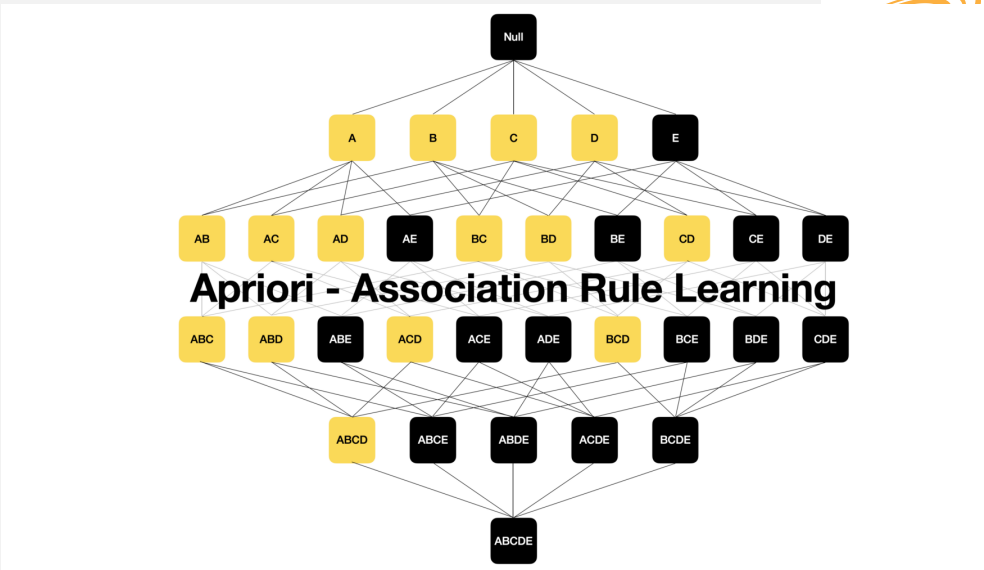


Figure 5: Generating Rules

Generating *Rules* for a Supported Itemset



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- If supported itemset $L \cup R$ has k elements, we can generate all the possible rules L $\rightarrow$ R systematically from it and then check the value of confidence for each one.
- To do this, it is only necessary to generate all possible right-hand sides in turn.
- Each one must have at least one and at most *k-1* elements.
- Having generated the right-hand side of a rule all the unused items in L $\cup$ R must then be on the left-hand side.
- For itemset {c, d, e} there are 6 possible rules that can be generated, as in figure 7.

Sample Transactions: Reminder

If a database comprises 8 transactions ($n = 8$) and there are 5 different items (unrealistically low), denoted by *a, b, c, d* and *e*, we have $m = 5$ and $I = \{a, b, c, d, e\}$:

Transaction number	Transactions (itemsets)
1	{a, b, c}
2	{a, b, c, d, e}
3	{b}
4	{c, d, e}
5	{c}
6	{b, c, d}
7	{c, d, e}
8	{c, e}

Figure 6: Sample Database of Transactions



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Only one of the rules has a confidence value greater than or equal to minconf (0.8)

Rule $L \rightarrow R$	count$(L \cup R)$	count(L)	confidence$(L \rightarrow R)$
$de \rightarrow c$	3	3	1.0
$ce \rightarrow d$	3	4	0.75
$cd \rightarrow e$	3	4	0.75
$e \rightarrow cd$	3	4	0.75
$d \rightarrow ce$	3	4	0.75
$c \rightarrow de$	3	7	0.43

Figure 7: Six Rules Generated



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Scaling Issues

- Assuming that *k* is reasonably small, say 10, this number is manageable.
- For $k = 10$ there are 2^{10} - 2 = 1,022 possible rules.
- However, as *k* becomes larger the number of possible rules rapidly increases.
- For $k = 20$, it is 1,048,574.
- Fortunately, we can reduce the number of candidate rules considerably using simple logic.



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Theorem 3

Transferring members of a supported itemset from the left-hand side of a rule to the right-hand side cannot increase the value of rule confidence.

Proof

For this purpose we will write the original rule as $A \cup B \rightarrow C$, where sets A, B and C all contain at least one element, have no elements in common and the union of the three sets is the supported itemset S.

Transferring the item or items in B from the left to the right-hand side then amounts to creating a new rule $A \rightarrow B \cup C$.

The union of the left- and right-hand sides is the same for both rules, namely the supported itemset S, so we have

$$\text{confidence}(A \rightarrow B \cup C) = \frac{\text{support}(S)}{\text{support}(A)}$$

$$\text{confidence}(A \cup B \rightarrow C) = \frac{\text{support}(S)}{\text{support}(A \cup B)}$$

It is clear that the proportion of transactions in the database matched by an itemset A must be at least as large as the proportion matched by a larger itemset $A \cup B$, i.e. $\text{support}(A) \geq \text{support}(A \cup B)$.

Hence it follows that $\text{confidence}(A \rightarrow B \cup C) \leq \text{confidence}(A \cup B \rightarrow C)$.



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Using Theorem 3

- If the confidence of a rule $\geq$ `minconf` is met, we will call the itemset on its righthand side *confident*.
- If not, we will call the right-hand itemset *unconfident*.
- From Theorem 3, we then have two important results that apply whenever the union of the itemsets on the two sides of a rule is fixed:
  - Any superset of an **unconfident** right-hand itemset is **unconfident**
  - Any (non-empty) subset of a **confident** right-hand itemset is **confident**



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Generating Rules: Conclusions

- This is very similar to the situation with supported itemsets described in *Apriori*.
- We can generate confident right-hand side itemsets of increasing cardinality in a way similar to *Apriori*, with a considerable reduction in the number of candidate rules for which the confidence needs to be calculated.
- If at any stage there are no more confident itemsets of a certain cardinality, there cannot be any of larger cardinality and the rule generation process can stop.



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Issues with Support and Confidence

- Despite considerable pruning, the number of rules with *support* and *confidence* greater than specified thresholds can be large.
- In some cases, they can generate bad rules.
- Assume: N=10,000 (total transactions); 6,000 included *Computer Games*; 7,500 included *Videos*; 4,000 included both.
- MinSup = 30% and MinConf = 60%.
- (*ComputerGames* $\rightarrow$ *Videos*) will have support (4000/10000) = 40% and confidence (4000/6000) = 66% and thus, is supported.
- In terms of probability, this is not a good rule as buying videos is actually 75%.
- In real terms, this is a *negative* correlation!



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Correlation Measures

- A **correlation measure** is used to augment the support-confidence framework for association rules.
- In addition to support and confidence, a rule is also assessed by the correlation between itemsets *A* and *B*.
- Ultimately, this correlation brings a measure of probability into the final decision.
- There are many different correlation measures from which to choose.



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The Lift Correlation Measure

- Lift is a simple correlation measure defined as:
- The occurrence of itemset A is independent of the occurrence of itemset B if $P(A \cup B) = P(A) \times P(B)$
- Otherwise, itemsets A and B are **dependent** and **correlated** as events.
- The **lift** of rule $L \cup R$ measures how many more times the items in L and R occur together in transactions than would be expected if the itemsets L and R were statistically independent.



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`lift(L` $\rightarrow R$) Definition

- The number of times the items in L and R occur together in transactions is just `count(`L $\cup$ R`)`.
- The number of times the items in L occur is `count(`L`)`.
- The *proportion* of transactions matched by R is `support(`R`)`.
- So if L and R were independent, the number of times L and R occur together is `count(L) X support(R)`.

$$lift(L \rightarrow R) = \frac{count(L \cup R)}{count(L)\ X\ support(R)} \quad (6)$$

Or *actual* $\div$ *predicted*



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`lift(L` $\rightarrow R$`)`: Analysis

- If the result is 1, there is no correlation: they are independent of each another.
- If the result < 1, then the occurrence of A is *negatively correlated* with the occurrence of B.
- If the result > 1, then the occurrence of A is *positively correlated* with the occurrence of B.
- This means that the occurrence of one **implies** the occurrence of the other.
- Or, the occurrence of one **lifts** the occurrence of the other.



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Calculating Lift: An Example

- Assume we have a database with 2000 transactions and a rule $L \rightarrow R$ with the following support counts
- `count(L) = 220; count(R) = 250; count(L` $\cup$ `R) = 190.`
- We can calculate `support(R) = count(R)/2000 = 0.125`

$$lift(L \rightarrow R) = \frac{190}{220\ X\ 0.125} = 6.91 \quad (7)$$

This means that purchasing the items in L make it 6.91 times more likely that the items in R are purchased (than would be expected).



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Interesting Rules

- Lift values greater than 1 are *interesting*.
- They indicate that transactions containing L tend to contain R more often than transactions that do not contain L.
- Although `lift` is a useful measure of interestingness, it is not always the best one to use.
- In some cases, a rule with *higher* **support** and *lower* **lift** can be more interesting than one with lower support and higher lift because it applies to more cases (L occurs more often).
- Another measure of interestingness that is sometimes used is *leverage*.



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Leverage

- **Leverage** measures the *difference* between the support for $L \cup R$ and the support that would be expected if L and R were independent.
- The former is just `support(L ∪ R)`.
- The frequencies (i.e. supports) of L and R are `support(L)` and `support(R)`, respectively.
- If L and R were independent, the expected frequency of both occurring in the same transaction would be the **product** of `support(L)` and `support(R)`.
- Example: If L and R both occurred in 50% of transactions, the expected frequency for both in the same transaction would be 25% (50% * 50%).

$$leverage(L \rightarrow R) = support(L \cup R) - support(L) \times support(R) \quad (8)$$

In other words: *Actual - Predicted*



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Calculating Leverage: An Example (1)

- The value of the **leverage** of a rule is *always less than* its support.
- The number of rules satisfying the `support ≥ minsup` and `confidence ≥ minconf` constraints can be reduced by setting a *leverage constraint*.
- For example, `leverage ≥ 0.0001`, corresponding to an improvement in support of one occurrence per 10,000 transactions in the database.
- Assume a database has 100,000 transactions and we have a rule $L \rightarrow R$ with these support counts
- `count(L) = 8000; count(R) = 9000; count (L ∪ R) = 7000`

$$leverage(L \rightarrow R) = 0.07 - (0.08 * 0.09) = 0.0628 \quad (9)$$



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Summary

- `support = 0.070 confidence = 0.875`
- `lift = 9.722 leverage = 0.063`
- Thus, the rule applies to 7% of the transactions in the database and is satisfied for 87.5% of the transactions that include the items in L.
- The actual occurrence is 9.722 times more frequent than would be expected by chance.
- The improvement in support compared with chance is 0.063, corresponding to 6.3 transactions per 100 in the database, i.e. approximately 6300 in the database of 100,000 transactions.



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